

Differential Equations Notes: Definitions, Separable Forms, Exact Equations & Integrating Factors

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1 Introduction & Basic Definitions (IMG_0983 – IMG_0985)

Introductory Word Problems

Problem 1: If Ram has x number of chocolates and I give him 3 chocolates, he will have 8 chocolates. How many chocolates did he originally have?

$$x + 3 = 8 \implies x = 8 - 3 \implies \mathbf{x = 5}$$

Problem 2 (Radioactive Decay): A radioactive element decays at a rate equal to the amount present left. Find the amount left after 10 minutes.

$$\frac{dy}{dx} = -y \implies \int \frac{dy}{y} = - \int dx \implies \ln y = -x \implies y = e^{-x}$$

For $x = 10$:

$$y(10) = e^{-10}$$

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Definitions: Differential Equations

An ****ordinary differential equation (ODE)**** is an equation involving derivatives of one dependent variable with respect to one or more independent variables.

$$y' = \cos x$$

$$\frac{d^2y}{dx^2} + 4y = 0$$

Order of a Differential Equation: The order of the highest derivative appearing in the differential equation is called the ****order**** of the differential equation.

Degree of a Differential Equation: The power of the highest-order derivative appearing in the differential equation is called the ****degree**** of the differential equation.

- $\frac{dy}{dx} = \cos x \implies$ Degree 1
- $\frac{d^2y}{dx^2} + 4y = 0 \implies$ Order 2, Degree 1

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Ordinary vs. Partial Differential Equations

- **ODE:** An equation involving derivatives of one dependent variable with respect to ****one**** independent variable.

$$y' = \cos x, \quad \frac{d^3 y}{dx^3} + 3x^2 \frac{dy}{dx} = 4y$$

- **PDE:** An equation involving derivatives of one dependent variable with respect to ****two or more**** independent variables.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

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Linearity & Homogeneity

Linear Differential Equation: A differential equation is said to be linear if:

1. Degree of the dependent variable and its derivatives is one.
2. There is no multiplication of dependent variable and its derivatives.

Examples:

- Linear: $\frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} + 5y = 0$
- Non-linear: $\left(\frac{d^2 y}{dx^2}\right)^2 + y^2 \frac{dy}{dx} + 5 = 0$

Homogeneity: A differential equation is said to be homogeneous if it has no constant/source term independent of y .

- Homogeneous: $y'' + 4y = 0$
- Non-homogeneous: $y'' + 4y = 5x^2$

Initial & Boundary Conditions: If the solution to a differential equation is obtained subject to conditions evaluated at the ****same point****, it is an ****Initial Value Problem (IVP)****. If evaluated at ****two or more points****, it is a ****Boundary Value Problem (BVP)****.

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2 Separable & Reducible Differential Equations (IMG_0986 – IMG_0988)

Separable Differential Equations

Form: $\frac{dy}{dx} = f(x)g(y) \implies \frac{dy}{g(y)} = f(x)dx$

Example 1:

$$(y + 1)dx = (x + 1)dy \implies \frac{dx}{x + 1} = \frac{dy}{y + 1}$$

$$\ln(y + 1) = \ln(x + 1) + \ln c \implies y + 1 = c(x + 1) \implies \mathbf{y} = \mathbf{c}(\mathbf{x} + \mathbf{1}) - \mathbf{1}$$

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Reducible to Separable Form (Homogeneous Substitution)

Example 2:

$$\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}$$

Substitute $y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$v + x \frac{dv}{dx} = \frac{x^3 v}{x^3(1 + v^3)} = \frac{v}{1 + v^3}$$

$$x \frac{dv}{dx} = \frac{v}{1 + v^3} - v = \frac{-v^4}{1 + v^3} \implies \frac{1 + v^3}{v^4} dv = -\frac{dx}{x}$$

Integrating both sides:

$$\int v^{-4} dv + \int \frac{1}{v} dv = -\int \frac{dx}{x} \implies -\frac{1}{3v^3} + \ln v = -\ln x + \ln c$$

$$-\frac{1}{3v^3} = \ln\left(\frac{c}{vx}\right) \implies \frac{1}{3v^3} = \ln\left(\frac{vx}{c}\right)$$

Since $v = \frac{y}{x}$:

$$\frac{x^3}{3y^3} = \ln\left(\frac{y}{c}\right) \implies \frac{y}{c} = e^{\frac{x^3}{3y^3}} \implies \mathbf{y} = \mathbf{c}e^{\frac{\mathbf{x}^3}{3\mathbf{y}^3}}$$

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Reducible to Homogeneous Form (Translation of Axes)

Example 3:

$$\frac{dy}{dx} = \frac{x - y + 1}{x + y + 3}$$

Let $x = X + h$ and $y = Y + k$:

$$\frac{dY}{dX} = \frac{X - Y + (h - k + 1)}{X + Y + (h + k + 3)}$$

Set constants to zero:

$$\begin{cases} h - k + 1 = 0 \\ h + k + 3 = 0 \end{cases} \implies h = -2, \quad k = 1$$

The reduced equation becomes:

$$\frac{dY}{dX} = \frac{X - Y}{X + Y}$$

Substitute $Y = vX \implies \frac{dY}{dX} = v + X \frac{dv}{dX}$:

$$v + X \frac{dv}{dX} = \frac{1-v}{1+v} \implies X \frac{dv}{dX} = \frac{1-v-v-v^2}{1+v} = \frac{1-2v-v^2}{1+v}$$

Separating variables:

$$\frac{1+v}{1-2v-v^2} dv = \frac{dX}{X}$$

Let $t = 1 - 2v - v^2 \implies dt = -2(1+v)dv$:

$$-\frac{1}{2} \int \frac{dt}{t} = \int \frac{dX}{X} \implies \ln t = -2 \ln X + \ln c \implies t = \frac{c}{X^2}$$

Substituting back $t = 1 - 2v - v^2$:

$$1 - 2v - v^2 = \frac{c}{X^2} \implies 1 - 2\left(\frac{Y}{X}\right) - \left(\frac{Y}{X}\right)^2 = \frac{c}{X^2}$$

Substituting $X = x + 2$ and $Y = y - 1$:

$$1 - 2\frac{(y-1)}{(x+2)} - \frac{(y-1)^2}{(x+2)^2} = \frac{c}{(x+2)^2}$$

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3 Exact Differential Equations & Integrating Factors (IMG_0989 – IMG_0992)

Exact Differential Equations

A differential equation $M(x, y)dx + N(x, y)dy = 0$ is ****exact**** if there exists a scalar function $f(x, y)$ such that:

$$\frac{\partial f}{\partial x} = M, \quad \frac{\partial f}{\partial y} = N$$

The necessary and sufficient condition for exactness is:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

General Solution Formula:

$$\int_{y=\text{const}} M(x, y)dx + \int (\text{terms in } N \text{ free from } x)dy = c$$

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Example 1: Solving an Exact Equation

$$(y \cos x + \sin y + y)dx + (\sin x + x \cos y + x)dy = 0$$

Here $M = y \cos x + \sin y + y$ and $N = \sin x + x \cos y + x$.

$$\frac{\partial M}{\partial y} = \cos x + \cos y + 1, \quad \frac{\partial N}{\partial x} = \cos x + \cos y + 1$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is ****exact****.

Solution:

$$\int (y \cos x + \sin y + y) dx + \int (0) dy = c$$
$$y \sin x + x \sin y + xy = c$$

Integrating Factors (I.F.)

If $Mdx + Ndy = 0$ is not exact, it can be made exact by multiplying by an ****Integrating Factor (I.F.)****.

Example 2 (Inspection Method):

$$ydx + (x^2y - x)dy = 0$$

Here $\frac{\partial M}{\partial y} = 1$ and $\frac{\partial N}{\partial x} = 2xy - 1$ (Not Exact). Multiplying the equation by $\frac{1}{x^2}$:

$$\frac{y}{x^2}dx + \left(y - \frac{1}{x}\right)dy = 0$$

Checking exactness:

$$\frac{\partial M'}{\partial y} = \frac{1}{x^2}, \quad \frac{\partial N'}{\partial x} = \frac{1}{x^2} \quad (\text{Exact})$$

Rules for Finding Integrating Factors (I.F.)

1. **Rule 1 (Homogeneous Form):** If $Mdx + Ndy = 0$ is homogeneous and $Mx + Ny \neq 0$, then:

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

2. **Rule 2 (Function of xy):** If equation is of the form $yf(xy)dx + xg(xy)dy = 0$ and $Mx - Ny \neq 0$, then:

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

3. **Rule 3 (Function of x alone):** If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$, then:

$$\text{I.F.} = e^{\int f(x)dx}$$

4. **Rule 4 (Function of y alone):** If $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y)$, then:

$$\text{I.F.} = e^{\int f(y)dy}$$

Practice Problems (IMG_0992)

1. Solve: $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$
2. Solve: $(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)xdy = 0$
3. Solve: $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$

4 First-Order Linear Differential Equations (IMG_0993 – IMG_0994)

Derivation of Integrating Factor (I.F.)

Standard Form:

$$y' + P(x)y = r(x)$$

This equation can be written as:

$$[P(x)y - r(x)]dx + dy = 0$$

This equation is not exact. Suppose $\mu(x)$ is an integrating factor such that:

$$\mu(x)[P(x)y - r(x)]dx + \mu(x)dy = 0 \quad \text{is exact.}$$

By the condition of exactness ($\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$):

$$\begin{aligned} \frac{\partial}{\partial y} (\mu(x)P(x)y - \mu(x)r(x)) &= \frac{\partial \mu(x)}{\partial x} \\ \implies \mu(x)P(x) &= \frac{d\mu}{dx} \implies \frac{d\mu}{\mu} = P(x)dx \implies \mu = e^{\int P(x)dx} \end{aligned}$$

After multiplying by the Integrating Factor (I.F.), the equation reduces to:

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int r(x) \cdot (\text{I.F.}) dx + c \\ \implies y &= (\text{I.F.})^{-1} \left[\int r(x) \cdot (\text{I.F.}) dx + c \right] \\ \mathbf{y} &= \mathbf{e}^{-\int \mathbf{P(x)}d\mathbf{x}} \left[\int \mathbf{r(x)}\mathbf{e}^{\int \mathbf{P(x)}d\mathbf{x}} d\mathbf{x} + \mathbf{c} \right] \end{aligned}$$

Example 1

Solve:

$$x \frac{dy}{dx} - (x+1)y = x^2 - x^3$$

Divide by x :

$$\frac{dy}{dx} - \left(1 + \frac{1}{x}\right)y = x - x^2$$

Here $P(x) = -\left(1 + \frac{1}{x}\right)$ and $r(x) = x - x^2$.

Integrating Factor:

$$\text{I.F.} = e^{\int P(x)dx} = e^{-\int(1+\frac{1}{x})dx} = e^{-(x+\ln x)} = e^{-x} \cdot e^{-\ln x} = \frac{x^{-1}e^{-x}}{1} = \frac{e^{-x}}{x}$$

General Solution:

$$y \cdot \left(\frac{e^{-x}}{x}\right) = \int (x - x^2) \left(\frac{e^{-x}}{x}\right) dx + c$$

$$yx^{-1}e^{-x} = \int x(1-x) \frac{e^{-x}}{x} dx = \int (1-x)e^{-x} dx$$

Integrating by parts:

$$yx^{-1}e^{-x} = [-e^{-x} - (-xe^{-x} - e^{-x})] + c$$

$$yx^{-1}e^{-x} = -e^{-x} + xe^{-x} + e^{-x} + c = xe^{-x} + c$$

Multiply both sides by xe^x :

$$y = xe^x[xe^{-x} + c]$$

$$\mathbf{y} = \mathbf{x}^2 + \mathbf{cxe}^{\mathbf{x}}$$

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5 Bernoulli's Differential Equation (IMG_0995 – IMG_0996)

Definition and Reduction to Linear Form

A differential equation of the form:

$$y' + P(x)y = g(x)y^a \quad (a \in \mathbb{R})$$

is called **Bernoulli's Equation**.

Reduction Procedure: Divide by y^a :

$$y^{-a}y' + P(x)y^{1-a} = g(x) \quad \text{--- (1)}$$

Let $t = y^{1-a}$. Differentiating with respect to x :

$$\frac{dt}{dx} = (1-a)y^{-a} \frac{dy}{dx} \implies y^{-a} \frac{dy}{dx} = \frac{1}{1-a} \frac{dt}{dx}$$

Substituting into (1):

$$\frac{1}{1-a} \frac{dt}{dx} + P(x)t = g(x)$$

$$\frac{dt}{dx} + (1-a)P(x)t = (1-a)g(x) \quad (\text{Linear in } t)$$

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Example 1

Solve:

$$\frac{dy}{dx} - \frac{y}{x} = \frac{y^2}{x} \ln x$$

Divide by y^2 :

$$y^{-2} \frac{dy}{dx} - \frac{y^{-1}}{x} = \frac{\ln x}{x}$$

$$\text{Let } t = y^{-1} \implies \frac{dt}{dx} = -y^{-2} \frac{dy}{dx} \implies y^{-2} \frac{dy}{dx} = -\frac{dt}{dx}.$$

$$-\frac{dt}{dx} - \frac{t}{x} = \frac{\ln x}{x} \implies \frac{dt}{dx} + \frac{1}{x}t = -\frac{\ln x}{x}$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

Solving for t :

$$t \cdot x = \int \left(-\frac{\ln x}{x} \right) x dx + c = - \int \ln x dx + c$$

$$tx = -(x \ln x - x) + c = -x \ln x + x + c$$

Divide by x :

$$t = x^{-1}(-x \ln x + x + c) = 1 - \ln x + cx^{-1}$$

Substitute back $t = y^{-1}$:

$$\mathbf{y^{-1} = 1 - \ln x + \frac{c}{x}}$$

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6 Applications: Electric Circuits (RL & RC Series Circuits) (IMG_0997 – IMG_0998)

Circuit Components & Governing Laws

- Resistor Voltage Drop: $V_R = IR$
- Inductor Voltage Drop: $V_L = L \frac{dI}{dt}$
- Capacitor Voltage Drop: $V_C = \frac{Q}{C}$

By **Kirchhoff's Voltage Law (KVL)***, the sum of voltage drops across components equals the applied EMF $E(t)$:

$$L \frac{dI}{dt} + RI + \frac{Q}{C} = E(t)$$

Since $I = \frac{dQ}{dt} = Q'$, we get the second-order DE:

$$LQ'' + RQ' + \frac{Q}{C} = E(t)$$

RL Circuit Equation:

$$L \frac{dI}{dt} + RI = E(t)$$

RC Circuit Equation:

$$R \frac{dQ}{dt} + \frac{Q}{C} = E(t)$$

Example 1 (RL Circuit)

Find the current at time $t > 0$ in a circuit having in series a constant EMF 40 V, a resistance 10Ω , and an inductance 0.2 H , given that initial current is zero ($I(0) = 0$).

Formulation:

$$R = 10 \Omega, \quad L = 0.2 \text{ H}, \quad E(t) = 40 \text{ V}, \quad I(0) = 0$$

$$L \frac{dI}{dt} + RI = E(t) \implies 0.2 \frac{dI}{dt} + 10I = 40$$

Divide by 0.2:

$$\frac{dI}{dt} + 50I = 200$$

Solving using I.F.:

$$\begin{aligned} \text{I.F.} &= e^{\int 50 dt} = e^{50t} \\ I \cdot e^{50t} &= \int 200e^{50t} dt + c = \frac{200}{50} e^{50t} + c = 4e^{50t} + c \\ I(t) &= 4 + ce^{-50t} \end{aligned}$$

Applying Initial Condition $I(0) = 0$:

$$0 = 4 + ce^0 \implies c = -4$$

$$\mathbf{I(t) = 4(1 - e^{-50t}) \text{ A}}$$

7 Orthogonal Trajectories (IMG_0999 & IMG_1001)

Definition & Procedure

An **orthogonal trajectory** is a curve that intersects each member of a family of curves at right angles (90°).

Method for Cartesian Coordinates:

1. Given family of curves: $f(x, y, c) = 0$.
2. Differentiate to find differential equation: $\frac{dy}{dx} = f(x, y)$.
3. Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$ to get differential equation of orthogonal trajectories:

$$\frac{dy}{dx} = -\frac{1}{f(x, y)}$$

4. Solve the resulting differential equation.
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Example 1: Parabolas $y = cx^2$

Find the orthogonal trajectories of the family of parabolas $y = cx^2$.

Step 1: Eliminate parameter c

$$y = cx^2 \implies c = \frac{y}{x^2}$$

Differentiate $y = cx^2$:

$$y' = 2cx = 2\left(\frac{y}{x^2}\right)x \implies \frac{dy}{dx} = \frac{2y}{x}$$

Step 2: Differential Equation of Orthogonal Trajectories

Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$:

$$-\frac{dx}{dy} = \frac{2y}{x} \implies \frac{dy}{dx} = -\frac{x}{2y}$$

Step 3: Solve by Separation of Variables

$$2ydy = -xdx \implies \int 2ydy = -\int xdx + c$$
$$y^2 = -\frac{x^2}{2} + c \implies \frac{x^2}{2} + y^2 = c \quad (\text{Family of Ellipses})$$

8 System of Linear Differential Equations (IMG_1002 – IMG_1003)

Matrix Form & Eigenvalue Method

A linear system of first-order differential equations:

$$\mathbf{y}' = A\mathbf{y}$$

General solution form:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 \mathbf{v}_2 e^{\lambda_2 t}$$

where λ_1, λ_2 are eigenvalues of A and $\mathbf{v}_1, \mathbf{v}_2$ are corresponding eigenvectors.

Example 1

Solve the system:

$$\begin{cases} \frac{dy_1}{dt} = 2y_1 + 2y_2 \\ \frac{dy_2}{dt} = 5y_1 - y_2 \end{cases}$$

Step 1: Write in Matrix Form

$$A = \begin{pmatrix} 2 & 2 \\ 5 & -1 \end{pmatrix}$$

Step 2: Find Eigenvalues $\det(A - \lambda I) = 0$

$$\begin{vmatrix} 2 - \lambda & 2 \\ 5 & -1 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(-1 - \lambda) - 10 = 0 \implies \lambda^2 - \lambda - 12 = 0$$

$$(\lambda + 3)(\lambda - 4) = 0 \implies \lambda_1 = -3, \quad \lambda_2 = 4$$

Step 3: Find Eigenvectors

Case 1: For $\lambda_1 = -3$

$$(A + 3I)\mathbf{v}_1 = \mathbf{0} \implies \begin{pmatrix} 5 & 2 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$5v_1 + 2v_2 = 0 \implies \mathbf{v}_1 = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

Case 2: For $\lambda_2 = 4$

$$(A - 4I)\mathbf{v}_2 = \mathbf{0} \implies \begin{pmatrix} -2 & 2 \\ 5 & -5 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-2v_1 + 2v_2 = 0 \implies v_1 = v_2 \implies \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 4: General Solution

$$\mathbf{y}(t) = \mathbf{c}_1 \begin{pmatrix} -2 \\ 5 \end{pmatrix} e^{-3t} + \mathbf{c}_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t}$$

Example 2: System of Linear ODEs with Repeated Eigenvalues

Given the system of first-order linear differential equations:

$$\begin{cases} \frac{dy_1}{dx} = -8y_1 - 2y_2 \\ \frac{dy_2}{dx} = 2y_1 - 4y_2 \end{cases}$$

Step 1: Matrix Formulation

The system can be written in matrix form $\mathbf{y}' = A\mathbf{y}$, where:

$$A = \begin{pmatrix} -8 & -2 \\ 2 & -4 \end{pmatrix}$$

Step 2: Characteristic Equation and Eigenvalues

To find the eigenvalues, set $\det(A - \lambda I) = 0$:

$$\begin{aligned} \begin{vmatrix} -8 - \lambda & -2 \\ 2 & -4 - \lambda \end{vmatrix} &= 0 \\ (-8 - \lambda)(-4 - \lambda) - (-4) &= 0 \\ \lambda^2 + 12\lambda + 32 + 4 = 0 &\implies \lambda^2 + 12\lambda + 36 = 0 \\ (\lambda + 6)^2 = 0 &\implies \lambda = -6, -6 \quad (\text{Repeated Eigenvalue}) \end{aligned}$$

Step 3: Finding the Eigenvector

For $\lambda = -6$, solve $(A + 6I)\mathbf{v} = \mathbf{0}$:

$$\begin{aligned} \begin{pmatrix} -8 + 6 & -2 \\ 2 & -4 + 6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} -2 & -2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{aligned}$$

From $-2v_1 - 2v_2 = 0 \implies v_2 = -v_1$. Choosing $v_1 = 1$:

$$\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Step 4: General Solution Setup for Repeated Eigenvalues

For a repeated eigenvalue λ with only one independent eigenvector $\mathbf{v}$, the general solution takes the form:

$$\mathbf{y}(t) = c_1 \mathbf{v} e^{\lambda t} + (c_2 \mathbf{v} t + \mathbf{u}) e^{\lambda t}$$

Substituting $\lambda = -6$ and $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$:

$$\begin{aligned} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} &= c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-6t} + \left(c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} t + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \right) e^{-6t} \\ \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} &= \begin{pmatrix} (c_1 + c_2 t + u_1) e^{-6t} \\ (-c_1 - c_2 t + u_2) e^{-6t} \end{pmatrix} \end{aligned}$$

Step 5: Solving for Generalized Eigenvector $\mathbf{u}$

Substituting $\mathbf{y}(t)$ back into $\mathbf{y}' = A\mathbf{y}$:

Left-Hand Side ($\mathbf{y}'$):

$$\frac{d}{dt} \begin{pmatrix} (c_1 + c_2 t + u_1)e^{-6t} \\ (-c_1 - c_2 t + u_2)e^{-6t} \end{pmatrix} = \begin{pmatrix} -6(c_1 + c_2 t + u_1) + c_2 \\ -6(-c_1 - c_2 t + u_2) - c_2 \end{pmatrix} e^{-6t}$$

Right-Hand Side ($A\mathbf{y}$):

$$\begin{pmatrix} -8 & -2 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} (c_1 + c_2 t + u_1)e^{-6t} \\ (-c_1 - c_2 t + u_2)e^{-6t} \end{pmatrix} \\ = \begin{pmatrix} -8(c_1 + c_2 t + u_1) - 2(-c_1 - c_2 t + u_2) \\ 2(c_1 + c_2 t + u_1) - 4(-c_1 - c_2 t + u_2) \end{pmatrix} e^{-6t}$$

Equating both sides leads to the generalized eigenvector equation $(A - \lambda I)\mathbf{u} = \mathbf{v}$:

$$\begin{pmatrix} -2 & -2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} c_2 \\ -c_2 \end{pmatrix}$$

Setting $c_2 = 1$ gives $-2u_1 - 2u_2 = 1$. A simple particular solution is:

$$u_1 = -\frac{1}{2}, \quad u_2 = 0$$